

Markov Text Generator

Nruta Choudhari, 2024

Below is the screenshot of the output of test_example.csv:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Markov Text Generation 1309/test_mtg.py"
input: she was not (n=2)
output: she was not be a very well , and the
expected output: she was not be a very well , and the
input: she was not (n=3)
output: she was not in the world , and the two
expected output: she was not in the world , and the two
input: i would ask her (n=3)
output: i would ask her reason , and the two miss
expected output: i would ask her reason , and the two miss
input: i would ask her (n=4)
output: i would ask her reason for thinking so , because
expected output: i would ask her reason for thinking so , because
input: they were sorry (n=4)
output: they were sorry to see them , and always openly
expected output: they were sorry to see them , and always openly
input: they were sorry (n=5)
output: they were sorry to see them .
expected output: they were sorry to see them .

[Done] exited with code=0 in 3.635 seconds
```

Below are some more examples to show for randomized and deterministic models.

1. n=2

```
Markov_chain_examples_NC.py > ...
5 corpus = [
6     "data", "analysis", "is", "important", "for", "making", "decisions",
7     "in", "business", "data", "can", "reveal", "trends", "and",
8     "patterns", "that", "help", "understand", "customer", "behavior",
9     "data", "analysis", "can", "also", "help", "in", "predictive",
10    "modeling", "and", "forecasting"
11 ]
12
13 initial_sentence = ["data", "analysis"]
14 n = 2
15
16 output1 = finish_sentence(initial_sentence, n, corpus, randomize=False)
17 print("the output for the corpus when n=2 and randomize = False")
18 print(" ".join(output1))
19 print("\n")
20 output2 = finish_sentence(initial_sentence, n, corpus, randomize=True)
21 print("the output for the corpus when n=2 and randomize = True")
22 print(" ".join(output2))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Markov Text Generation 1309/tempCodeRunnerFile.py"
the output for the corpus when n=2 and randomize = False
data analysis can also help in business data analysis can

the output for the corpus when n=2 and randomize = True
data analysis is important for making decisions in business data

[Done] exited with code=0 in 0.39 seconds
```

2. $n = 3$

```
test_mtg.py • Markov_chain_examples_NC.py •
Markov_chain_examples_NC.py > ...
26 corpus = [
27     "in", "the", "heart", "of", "the", "city", "there",
28     "is", "a", "small", "cafe", "where", "people",
29     "gather", "to", "drink", "coffee", "and", "chat",
30     "about", "life", "and", "share", "their", "stories",
31     "it", "is", "a", "place", "of", "comfort"
32 ]
33
34
35 initial_sentence = ["in", "the", "heart"]
36 n = 3
37
38 output1 = finish_sentence(initial_sentence, n, corpus, randomize=False)
39 print("the output for the corpus when n=3 and randomize = False")
40 print(" ".join(output1))
41 print("\n")
42 output2 = finish_sentence(initial_sentence, n, corpus, randomize=True)
43 print("the output for the corpus when n=3 and randomize = True")
44 print(" ".join(output2))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Markov Text Generation 1309/tempCodeRunnerFile.py"

the output for the corpus when n=3 and randomize = False
in the heart of the city there is a place

the output for the corpus when n=3 and randomize = True
in the heart of the city there is a small

[Done] exited with code=0 in 0.401 seconds

3. $n=4$

```
test_mtg.py • Markov_chain_examples_NC.py •
Markov_chain_examples_NC.py > ...
45
46 corpus = [
47     "data", "science", "is", "a", "field", "that", "uses", "data",
48     "data", "science", "methods", "to", "analyze", "data", "data",
49     "science", "helps", "in", "making", "decisions", "based", "on",
50     "data", "data", "science", "can", "reveal", "patterns", "patterns",
51     "help", "in", "understanding", "the", "data", "data", "science",
52     "is", "used", "to", "improve", "business", "strategies", "using",
53     "data", "data", "science", "provides", "insights", "that", "can",
54     "drive", "innovation", "data", "science", "is", "essential", "for",
55     "making", "informed", "decisions", "across", "various", "industries"
56 ]
57
58 initial_sentence = ["data", "science"]
59
60 n = 4
61 |
62 output1 = finish_sentence(initial_sentence, n, corpus, randomize=False)
63 print("the output for the corpus when n=4 and randomize = False")
64 print(" ".join(output1))
65 print("\n")
66 output2 = finish_sentence(initial_sentence, n, corpus, randomize=True)
67 print("the output for the corpus when n=4 and randomize = True")
68 print(" ".join(output2))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Markov Text Generation 1309/tempCodeRunnerFile.py"
the output for the corpus when n=4 and randomize = False
data science is a field that uses data data science

the output for the corpus when n=4 and randomize = True
data science is used to improve business strategies using data

[Done] exited with code=0 in 0.338 seconds
```